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Models for Third Down Conversion in the National Football League

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Abstract

Several models are proposed for the probability of converting a third down attempt in the National Football League. The probability, which can depend on the number of yards to go, the strength of the offense, and the strength of the defense, leads to a logistic regression. We approach the problem through a hierarchical Bayes model and estimate parameters by using Markov chain Monte Carlo (MCMC). This MCMC estimation in the context of a hierarchical Bayes model may be relevant in other sports situations where a probability depends on the difference of strengths of the two teams. We find that the statistic "third-down conversion rate" to be a nearly meaningless measure of the efficiency of an offense. Even when this is adjusted for yards to go for a first down, there is little evidence that teams differ in their ability to achieve a first down on a third down conversion.

KEYWORDS: hierarchical Bayes, Markov chain Monte Carlo

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1. Introduction and Background

American football began in the 1870s as a continuous flow game similar to rugby. Players could carry the ball, pass the ball (backwards or sideways), or kick the ball. If a player was tackled with the ball, a scrum (a sort of “face off” with both teams arranged in a tight pattern) ensued. A player who carried the ball across the goal line got a chance (against no opposition) to kick the ball through the goal posts. This free flowing game changed in 1880 (see Carroll, Palmer and Thorne, 1998) when Walter Camp introduced rules that would allow one team to maintain possession. After a player was tackled, his team was allowed to kick the ball (not “hike”) from the point of the tackle to another player on his team. Thus, the scrum was replaced by the “line of scrimmage.” There was no limit to the number of times that a team could repeat this process. Weak teams were therefore able to hold on to the ball (much like the “four corners” offense in college basketball before the introduction of the shot clock) and run the clock out. The next year, the rules were changed to give teams just three chances (or downs) to have the ball. If the team gained five yards or lost ten yards from the original line of scrimmage, they got a new set of downs, or a “first down” in modern language. In 1910 a fourth down was added.

Current rules allow a team four downs to advance the ball ten yards, and if this is achieved, the team gets a new set of four downs (i.e., a *first down*). If a team fails to advance the ball ten yards, the opposing team takes over possession at the current line of scrimmage. On fourth down (the last chance to get a first down), most teams choose to punt (kick) the ball to the other team. This often puts the ball 35 to 40 yards down the field where the other team takes possession. Punting is done to avoid the risk of not making a first down on a fourth down attempt and giving the ball to the other team in a much better position (from the view of the opposing team). Therefore, third down is usually the last chance a team has to make a first down.

Data are kept and reported on the number of third down plays and the number of conversions. We often hear during an NFL telecast statistics about what percentage of times a team has been successful on their third-down plays. For example, we might hear that one team is 0 for 4, while their opponent is 3 for 5. Since the likelihood of making a first down on a third down play depends on the number of yards to go on third down, these raw statistics can be misleading. In the example stated, the team that was 0 for 4 might have had several penalties and incomplete passes early in their set of downs, leading to third-and-long every time. On the other hand, the team that was 3 for 5 might have been successful gaining a few yards by running or passing on first

and second down, setting up third and short on most of their third down plays. It is clear then, that the raw percentage of successful third-down attempts is not a good measure of the team's play on third down.

One might think that over the course of a season, the yards to go (YTG) on third down will average out. There is, in fact, quite a bit of variability in the distribution of YTG across the 32 NFL teams. Figure 1 shows histograms for the YTG for all 32 teams for the 2007 season. The average YTG varied from 5.8 for the Saints to 7.7 for the Rams. For many teams, the number of yards to go on third down is roughly uniform over the interval from 1 to 10, with probabilities tailing off after 10 yards. Teams facing fewer YTG on average will tend to have a higher conversion rate for this reason alone. For example, the Saints had the lowest average YTG on third down (5.8) and the highest raw third down conversion rate (56.7%) and the Rams had the highest YTG on third down (7.7) and the lowest raw conversion rate (28.1%). Table 1 shows the 2007 NFL standings, the raw third-down conversion rates, and other statistics for the season.

Even if the teams' yards to go on third down averaged about the same, comparing the percentage of times that teams convert third down plays is still troublesome because some teams play in a division that is loaded with strong defensive teams. A team that makes 30% of first downs in a division with weak defensive teams will not be as strong as a team that makes 30% of its first downs in a division with stronger defensive teams.

In order to compare the strengths of offenses in converting third down plays adjusted for YTG (and in the case of Model 3, the strengths of defensive teams in *stopping* third down plays) we propose three models for the probability of a successful a third-down conversion.

Model 1 assumes that the probability depends on YTG according to a logistic regression model, but does not depend on the offensive or defensive teams.

Model 2 assumes that the probability depends on YTG and which team is on offense.

Model 3 assumes that the probability depends on YTG and both the offensive and defensive teams.

For Model 1, we propose a Bayesian model for the parameters in the logistic regression. For Models 2 and 3, we propose a hierarchical Bayes model for the team-specific parameters. A hierarchical Bayes approach borrows information across all other teams in the sense that all other teams influence

Figure 1: Histograms of yards to go on third down. There were a few instances of more than thirty yards to go (Bears 3rd and 37, Chargers 3rd and 31 and 3rd and 32, Giants 3rd and 33, Raiders 3rd and 31 and 3rd and 40, Seahawks 3rd and 40, Steelers 3rd and 31, and Texans 3rd and 35). The number within each plot is the average number of yards to go.

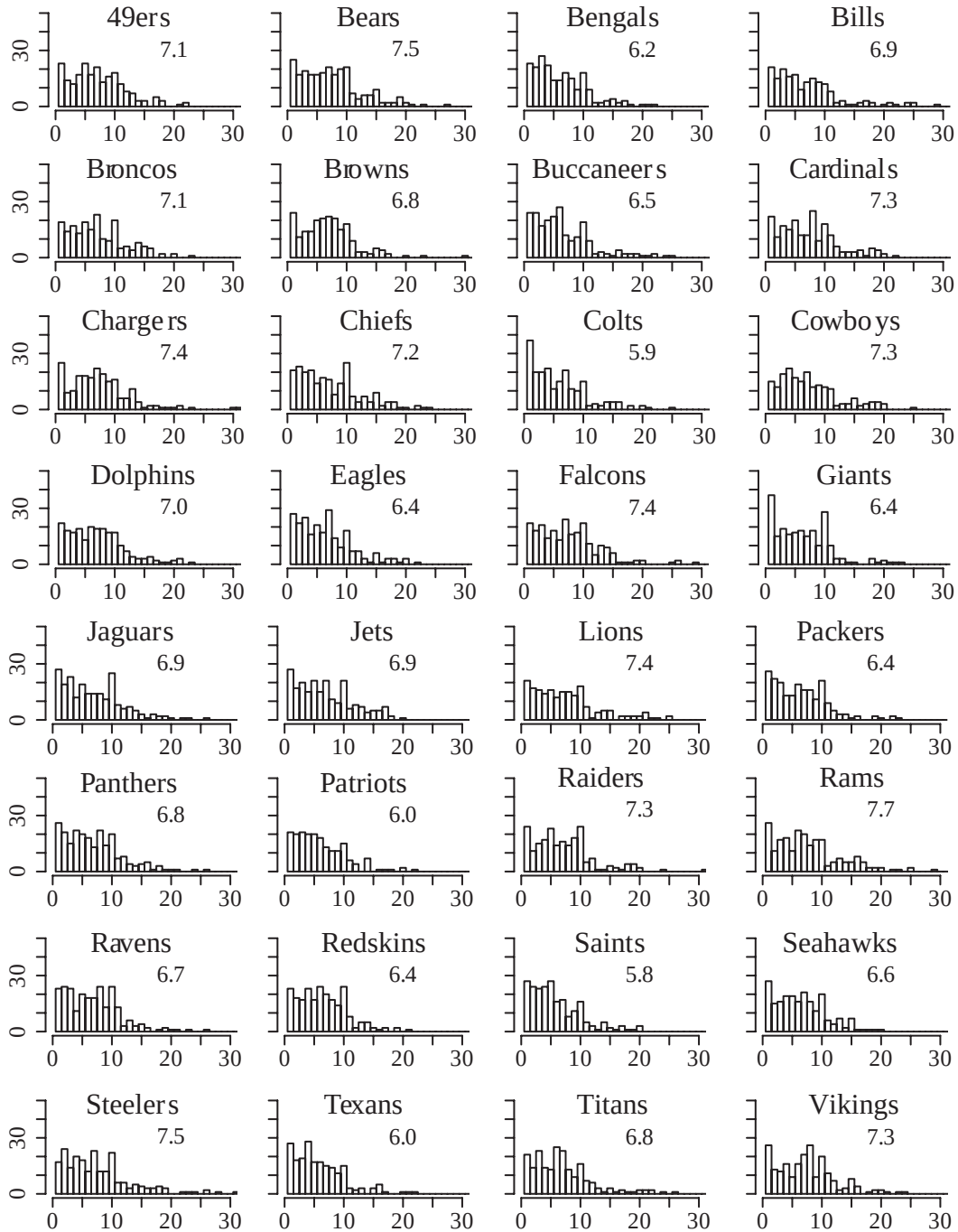


Table 1: Standings, Points, Yards, Turnovers and Raw Third Down Conversion Rates for the 2007 NFL Season

												Raw 3rd Down
AFC East	W	L	Points			Yards			Turnovers			Conv Rate
			For	Against	Diff	For	Against	Diff	For	Against	Diff	
New England Patriots	16	0	589	274	315	6580	4613	1967	31	15	16	45.9
Buffalo Bills	7	9	252	354	−102	4434	5807	−1373	30	21	9	32.5
New York Jets	4	12	268	355	−87	4715	5310	−595	21	25	−4	34.7
Miami Dolphins	1	15	267	437	−170	4600	5475	−875	22	29	−7	33.3
AFC North												
Pittsburgh Steelers	10	6	393	269	124	5239	4262	977	25	22	3	45.9
Cleveland Browns	10	6	402	382	20	5621	5753	−132	27	29	−2	38.7
Cincinnati Bengals	7	9	380	385	−5	5568	5580	−12	35	30	5	36.5
Baltimore Ravens	5	11	275	384	−109	4832	4825	7	23	40	−17	42.4
AFC South												
Indianapolis Colts	13	3	450	262	188	5739	4475	1264	37	19	18	34.6
Jacksonville Jaguars	11	5	411	304	107	5719	5021	698	30	21	9	33.6
Tennessee Titans	10	6	301	297	4	4987	4665	322	34	34	0	36.7
Houston Texans	8	8	379	384	−5	5337	5507	−170	25	38	−13	41.5
AFC West												
San Diego Chargers	11	5	412	284	128	5044	5124	−80	48	24	24	48.8
Denver Broncos	7	9	320	409	−89	5541	5376	165	30	29	1	30.8
Oakland Raiders	4	12	283	398	−115	4717	5466	−749	26	37	−11	35.7
Kansas City Chiefs	4	12	226	335	−109	4429	5111	−682	22	33	−11	36.0
												Raw 3rd Down
NFC East	W	L	Points			Yards			Turnovers			Conv Rate
Dallas Cowboys	13	3	455	325	130	5851	4922	929	29	24	5	39.4
New York Giants	10	6	373	351	22	5302	4880	422	25	34	−9	37.4
Washington Redskins	9	7	334	310	24	5334	4884	450	24	29	−5	37.0
Philadelphia Eagles	8	8	336	300	36	5729	4982	747	19	27	−8	41.5
NFC North												
Green Bay Packers	13	3	435	291	144	5931	5013	918	28	24	4	48.1
Minnesota Vikings	8	8	365	311	54	5379	5410	−31	31	30	1	38.4
Detroit Lions	7	9	346	444	−98	5166	6042	−876	35	36	−1	35.8
Chicago Bears	7	9	334	348	−14	4692	5675	−983	33	34	−1	32.5
NFC South												
Tampa Bay Bucs	9	7	334	270	64	5229	4454	775	35	20	15	36.9
Carolina Panthers	7	9	267	347	−80	4559	5197	−638	30	29	1	40.5
New Orleans Saints	7	9	379	388	−9	5780	5570	210	23	30	−7	56.7
Atlanta Falcons	4	12	259	414	−155	4816	5688	−872	28	24	4	44.2
NFC West												
Seattle Seahawks	10	6	393	291	102	5583	5149	434	34	24	10	33.8
Arizona Cardinals	8	8	404	399	5	5505	5283	222	29	36	−7	32.2
San Francisco 49ers	5	11	219	364	−145	3797	5539	−1742	22	34	−12	29.4
St. Louis Rams	3	13	263	438	−175	4760	5457	−697	27	37	−10	28.1

the estimates of the hyperparameters, which in turn affect the estimates of parameters from the individual teams. With many parameters to estimate, a frequentist approach is likely to produce estimates that are too disperse, whereas a hierarchical Bayes method will tend to shrink the estimates toward the middle. The assumption that the teams vary in their ability to do any task makes sense. For example, if team i 's "true" ability on some task is θ_i , and if the abilities vary across all teams, then it seems reasonable, that $\theta_1, \theta_2, \dots, \theta_{32}$ have some common prior distribution. This is similar to a random effects model in ANOVA, where the treatment means are assumed to have come from some distribution. This leads to the concept of exchangeability as discussed in Albert and Chib (1997).

Section 2 describes the logistic model and how it accounts for YTG and the teams on offense and defense. Sections 3, 4, and 5 describe, respectively, Models 1, 2, and 3. For Models 2 and 3 we describe a hierarchical Bayes model and we use Markov chain Monte Carlo (MCMC) to estimate the parameters. This MCMC estimation in the context of a hierarchical Bayes model may be relevant in other sports situations where a probability depends on the difference of strengths of the two teams or players.

2. The Logistic Model

We assume that the probability π of making a first-down depends on YTG, which we call x , in the logistic form

$$\pi = \frac{\exp[\alpha + \beta x]}{1 + \exp[\alpha + \beta x]} \quad (1)$$

or, in terms of the logit, or log odds transformation,

$$\text{logit } \pi = \log\left(\frac{\pi}{1 - \pi}\right) = \alpha + \beta x. \quad (2)$$

We can make various assumptions about the parameters α and β . The simplest model, of course, is to assume that α and β are the same for every third down play, regardless of the offensive or defensive team. This is Model 1, and is analyzed in the next section.

Model 2 assumes that the parameters α and β depend on the offensive team, but not on the defensive team. In this case, the probability that team i converts a third-down-and- x -yards-to-go is

$$\pi_i = \frac{\exp[\alpha_i + \beta_i x]}{1 + \exp[\alpha_i + \beta_i x]}, \quad i = 1, 2, \dots, 32. \quad (3)$$

Here, α_i and β_i represent the slope and intercept for logit π for team i .

The most thorough model (Model 3) assumes different α parameters and different β parameters for the offensive and defensive teams. Under this model, the parameters α and β in (1) and (2) depend on the difference between the strength of the offense and defense; specifically, $\alpha = \alpha_{Oi} - \alpha_{Dj}$ and $\beta = \beta_{Oi} - \beta_{Dj}$. The logit is thus

$$\text{logit } \pi_{ij} = (\alpha_{Oi} - \alpha_{Dj}) + (\beta_{Oi} - \beta_{Dj}) x \quad (4)$$

which makes the probability of conversion equal to

$$\pi_{ij} = \frac{\exp [(\alpha_{Oi} - \alpha_{Dj}) + (\beta_{Oi} - \beta_{Dj}) x]}{1 + \exp [(\alpha_{Oi} - \alpha_{Dj}) + (\beta_{Oi} - \beta_{Dj}) x]}. \quad (5)$$

The differencing in (5) creates an identifiability problem, because adding any fixed constant to both α_{Oi} and α_{Dj} (or to β_{Oi} and β_{Dj}) produces exactly the same probability. With a hierarchical Bayes method, we can overcome this identifiability problem by “anchoring” the prior distribution for two of the parameters at 0. We have assumed that the prior distributions for the defensive parameters α_{Dj} and β_{Dj} have mean zero (although we could just as easily have done this for the offensive parameters instead). With the defensive parameters having a prior mean of zero, we can then interpret the parameters α_{Oi} , β_{Oi} , α_{Dj} , and β_{Dj} in (5) from the perspective of offensive team i . The probability of making a first down involves the parameters α_{Oi} and β_{Oi} , just as with any first-order logistic regression, but these values are adjusted up or down according to the strength of the defense that team i is facing at the time.

If we knew the strengths of the defenses, i.e., the α_{Dj} and β_{Dj} parameters, then we could use the method of maximum likelihood or a straightforward Bayesian approach to estimate the offensive parameters α_{Oi} and β_{Oi} ; conversely, if we knew the offensive parameters, we could estimate the defensive parameters. The problem is that *neither* is known. This is similar to the problem in educational measurement, where we often want to estimate the ability of a set of examinees and simultaneously the difficulty of the test items; in this situation, the probability of getting a test item correct is a function of the difference between the examinee’s ability and the item difficulty. This problem has been addressed by Bock and Aitken (1981) and Rigdon and Tsutakawa (1983), who applied the EM algorithm of Dempster, Laird, and Rubin (1977), although Albert and Chib (1997) suggested a more efficient, and more generally applicable, method using Markov chain Monte Carlo (MCMC). We will address the use of MCMC in the next section. This model is related to

a Bradley-Terry (1952) model since the log-odds of success involves a difference between parameters for the offense and the defense. Bock (1997) pointed out the connection between the educational measurement problem and the Bradley-Terry model.

We analyze data from the 2007 NFL season. For each third down attempt, we recorded the offensive team, the defensive team, the yards to go, and whether a first-down was achieved. Define

$$Y_{ijk} = \begin{cases} 1, & \text{if } k\text{th attempt of offensive team } i \text{ on defensive team } j \text{ is successful} \\ 0, & \text{otherwise} \end{cases}$$

and

x_{ijk} = number of yards needed to make a first down.

Then, Y_{ijk} has a Bernoulli distribution (i.e., a binomial distribution with $n = 1$) having probability of success π_{ijk} . Under Model 3, this is

$$\pi_{ijk} = \frac{\exp [(\alpha_{Oi} - \alpha_{Dj}) + (\beta_{Oi} - \beta_{Dj}) x_{ijk}]}{1 + \exp [(\alpha_{Oi} - \alpha_{Dj}) + (\beta_{Oi} - \beta_{Dj}) x_{ijk}]}, \quad (6)$$

giving altogether 128 unknown model parameters that must be estimated: 32 values (one for each NFL team) for α_{Oi} , 32 values for α_{Dj} , 32 values for β_{Oi} , and 32 values for β_{Dj} .

3. Model 1: Conversion Probability Curve is the Same for All Teams

The simplest model assumes that all teams have the same logistic curve for the probability of making a first down on a third down attempt. In this case, the probability is just

$$\pi_{ijk} = \frac{\exp (\alpha + \beta x_{ijk})}{1 + \exp (\alpha + \beta x_{ijk})}.$$

There are just two parameters to estimate: α and β . For Model 1, we assume very diffuse independent priors: $\alpha \sim N(0, 1)$, and $\beta \sim N(0, 1)$. With such a large sample, the prior has little effect on the posterior distribution.

We then used the software WinBUGS (Lunn et al. (2000)) which uses Gibbs sampling to estimate the posterior distributions through simulation. Basically, Gibbs sampling simulates through several thousand iterations of a Markov process whose steady state distribution is the desired posterior distribution. Thus, we need to simulate until the steady state is essentially reached, and then simulate enough additional observations (from the steady state) that we

can get a good idea about the important characteristics, such as the posterior mean and standard deviation. The simulations made before we infer that the steady state is reached is called the burn-in. In Appendix A we discuss the issue of convergence to the steady state. We have found that several hundred to a few thousand iterations is generally sufficient for the burn-in period. We thus discard the first 10,000 iterations, and then run an additional 100,000 iterations (110,000 in total). We do this for four separate starting points, giving a total of 400,000 observations from which we estimate the posterior distributions. The parameter estimates for this model are shown in Table 2, and the estimated logistic curve is shown in Figure 2.

Table 2: Parameter estimates α and β in Model 1.

Parameter	Posterior	Posterior
	Mean	St. Dev.
α	0.6448	0.04724
β	-0.1529	0.00654

As expected, the estimate for β is negative, indicating that as YTG increases, the probability of conversion decreases. If $\alpha + \beta x$ is fairly close to zero, then

$$\frac{d}{dx} \left[\frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)} \right] = \beta \frac{\exp(\alpha + \beta x)}{[1 + \exp(\alpha + \beta x)]^2} \approx \beta \frac{e^0}{(1 + e^0)^2} = \frac{\beta}{4}$$

so that each additional YTG decreases the probability of successful conversion by approximately $0.1529/4 \approx 0.04$.

4. Model 2: Separate Curves for Each Offensive Team

Model 2 assumes that the conversion probability depends on YTG and on the offensive team, but not on the defensive team. We assume that the model parameters (the α 's and β 's) have prior distributions

$$\begin{aligned} \alpha_1, \alpha_2, \dots, \alpha_{32} &\sim \text{i.i.d. } N(\mu_1, \tau_1^{-1}), \text{ and} \\ \beta_1, \beta_2, \dots, \beta_{32} &\sim \text{i.i.d. } N(\mu_2, \tau_3^{-1}). \end{aligned}$$

Here, the τ 's represent the prior precision, that is, the reciprocal of the variance. (The reason for using τ_3 instead of τ_2 will become apparent in the next section.) The parameters μ_1 , μ_2 , τ_1 , and τ_3 are called hyperparameters, and each has its own prior distribution whose parameters are known. (If we assumed these parameters were unknown, we would have to put priors on these,

Figure 2: Probability of first down conversion assuming all teams have the same offensive and defensive strength (Model 1)

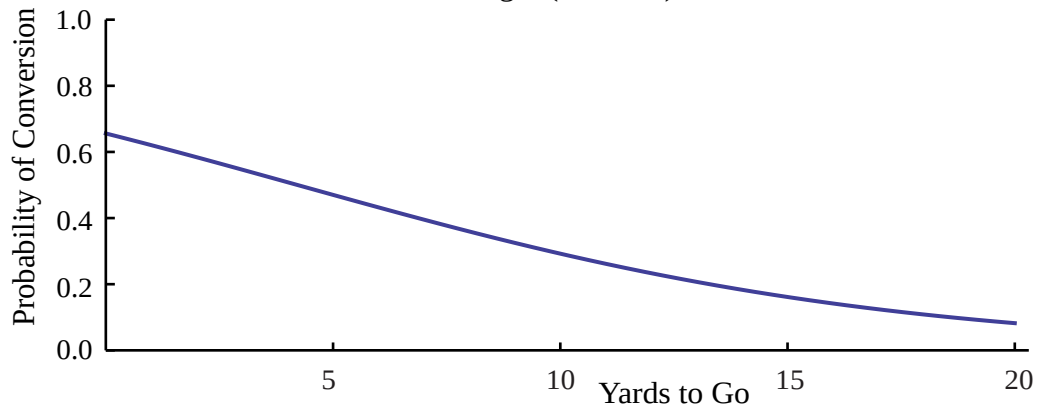


Figure 3: Probability of first down conversion on offense for each of the 32 teams against an average defense under Model 3.

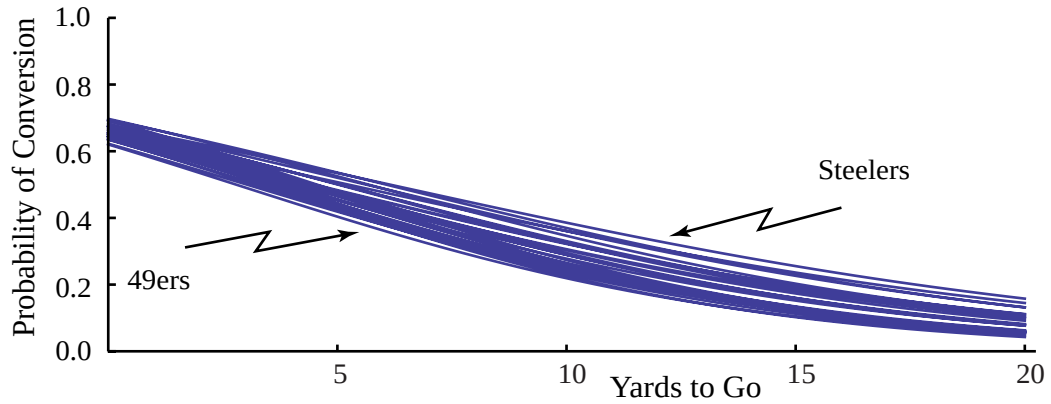
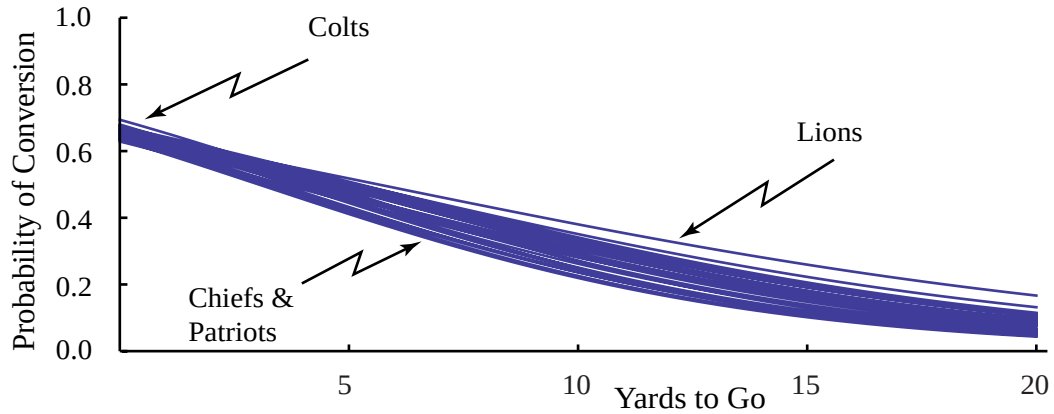


Figure 4: Probability of allowing a first down conversion by offense for each of the 32 teams against an average offense under Model 3.



and the process would never end.) The choice of prior distributions for Model 3 is detailed in Appendix B. We use the same priors for Model 2 as we do for Model 3, although Model 3 has more parameters. The prior distributions for μ_1 , μ_2 , τ_1 , and τ_3 are

$$\begin{aligned}\mu_1 &\sim N\left(0.7, \frac{1}{25}\right), & \tau_1 &\sim \text{GAMMA}\left(1, \frac{1}{8}\right), \\ \mu_2 &\sim N\left(-0.16, \frac{1}{1000}\right), & \tau_3 &\sim \text{GAMMA}\left(1, \frac{1}{100}\right).\end{aligned}$$

Point estimates of parameters are taken to be the posterior means. Estimates of the hyperparameters are shown in Table 3 and estimates for the model parameters (32 α 's and 32 β 's) are shown in Table 4.

Table 3: Estimates of hyperparameters for Model 2.

Parameter	Posterior Mean	Posterior St. Dev.	Parameter	Posterior Mean	Posterior St. Dev.
μ_1	0.6497	0.05553	τ_1	31.61	12.10
μ_2	-0.1533	0.008891	τ_3	776.4	223.8

5. Model 3: A Hierarchical Bayes Model with Parameters for Each Offensive and Defensive Team

We consider a model with two parameters (α_{Oi} and β_{Oi}) for each offense and two parameters (α_{Dj} and β_{Dj}) for each defense. To estimate the 128 parameters (32 values for each of α_{Oi} , α_{Dj} , β_{Oi} , β_{Dj}), we take a hierarchical Bayes approach and assume priors

$$\begin{aligned}\alpha_{O,1}, \alpha_{O,2}, \dots, \alpha_{O,32} &\sim \text{i.i.d. } N(\mu_1, \tau_1^{-1}), \\ \alpha_{D,1}, \alpha_{D,2}, \dots, \alpha_{D,32} &\sim \text{i.i.d. } N(0, \tau_2^{-1}), \\ \beta_{O,1}, \beta_{O,2}, \dots, \beta_{O,32} &\sim \text{i.i.d. } N(\mu_2, \tau_3^{-1}), \text{ and} \\ \beta_{D,1}, \beta_{D,2}, \dots, \beta_{D,32} &\sim \text{i.i.d. } N(0, \tau_4^{-1}).\end{aligned}$$

Here the τ 's represent the precision of the normal distribution, that is, the reciprocal of the variance. The parameters $\mu_1, \mu_2, \tau_1, \tau_2, \tau_3$, and τ_4 are hyperparameters, and each has its own prior distribution which is known completely. Note that the defensive parameters have a prior centered at 0, whereas the offensive parameters α_{Oi} and β_{Oi} have priors that are centered at arbitrary values: μ_1 and μ_2 , respectively. Altogether, there are 134 parameters to estimate: $4 \times 32 = 128$ parameters in the logistic model, plus the six hyperparameters.

Table 4: Estimated parameters α_O and β_O for Model 2.

Team	$\alpha_{O,i}$		$\beta_{O,i}$	
	Posterior Mean	Posterior StDev	Posterior Mean	Posterior StDev
49ers	0.2309	0.0323	-0.1697	0.0230
Bears	0.2400	0.0318	-0.1791	0.0221
Bengals	0.3582	0.0409	-0.1305	0.0223
Bills	0.2439	0.0370	-0.1630	0.0243
Broncos	0.3059	0.0371	-0.1436	0.0219
Browns	0.3242	0.0379	-0.1468	0.0223
Buccaneers	0.2624	0.0363	-0.1690	0.0234
Cardinals	0.2742	0.0353	-0.1652	0.0222
Chargers	0.3194	0.0365	-0.1336	0.0215
Chiefs	0.2429	0.0329	-0.1695	0.0223
Colts	0.3640	0.0423	-0.1373	0.0226
Cowboys	0.3300	0.0380	-0.1439	0.0216
Dolphins	0.2807	0.0351	-0.1607	0.0220
Eagles	0.2723	0.0358	-0.1780	0.0231
Falcons	0.2745	0.0330	-0.1568	0.0213
Giants	0.2761	0.0369	-0.1767	0.0233
Jaguars	0.3570	0.0378	-0.1229	0.0209
Jets	0.2971	0.0355	-0.1412	0.0216
Lions	0.2584	0.0358	-0.1646	0.0227
Packers	0.3262	0.0394	-0.1326	0.0225
Panthers	0.2522	0.0338	-0.1692	0.0226
Patriots	0.3494	0.0430	-0.1451	0.0237
Raiders	0.2949	0.0365	-0.1529	0.0221
Rams	0.2871	0.0342	-0.1475	0.0210
Ravens	0.2622	0.0339	-0.1645	0.0224
Redskins	0.2954	0.0366	-0.1492	0.0226
Saints	0.3329	0.0409	-0.1416	0.0230
Seahawks	0.2508	0.0350	-0.1639	0.0233
Steelers	0.3887	0.0372	-0.1178	0.0199
Texans	0.2679	0.0393	-0.1771	0.0248
Titans	0.2979	0.0380	-0.1565	0.0228
Vikings	0.2635	0.0343	-0.1668	0.0223

The prior distributions for the parameters were based on data from the 2003 season (Rigdon and Rigdon, 2004) and were selected so that any reasonable logistic curve has a chance of occurring *a priori*. If anything, we err on the side of having priors that are too diffuse, rather than too tight. We chose the following priors for the hyperparameters; the explanation of how these prior parameters were obtained is a bit involved, so we put the explanation in Appendix B. The priors are

$$\begin{aligned}\mu_1 &\sim N\left(0.7, \frac{1}{25}\right), & \tau_1 &\sim \text{GAMMA}\left(1, \frac{1}{8}\right), & \tau_2 &\sim \text{GAMMA}\left(1, \frac{1}{8}\right), \\ \mu_2 &\sim N\left(-0.16, \frac{1}{1000}\right), & \tau_3 &\sim \text{GAMMA}\left(1, \frac{1}{100}\right), & \tau_4 &\sim \text{GAMMA}\left(1, \frac{1}{100}\right).\end{aligned}$$

For a reader interested in just whether these priors are broad enough to cover any reasonable logistic curve, we suggest taking a look at Figure 9, where we plot one hundred logistic curves whose parameters were chosen at random from these prior distributions. Although there are some clearly unreasonable curves, any reasonable curve should be covered by these priors.

Again, we used WinBUGS to estimate the posterior distributions. The estimates for the six hyperparameters ($\mu_1, \mu_2, \tau_1, \tau_2, \tau_3, \tau_4$) are shown in Table 5. Table 6 shows the posterior mean and standard deviation of all $4 \times 32 = 128$ of the model parameters for Model 3. The posterior means for the defensive “intercept” parameters and the defensive “slope” parameters are centered very close to zero, which is to be expected since the prior mean was 0.

Table 5: Estimates of hyperparameters for Model 3.

Parameter	Posterior Mean	Posterior St. Dev.	Parameter	Posterior Mean	Posterior St. Dev.
μ_1	0.6614	0.06279	τ_1	30.88	11.93
μ_2	-0.1548	0.01097	τ_2	37.60	13.73
			τ_3	728.90	213.30
			τ_4	734.00	214.70

All 32 logistic curves for offensive teams playing against an average defensive team under Model 3 are shown in Figure 3. The curve that is lowest (almost uniformly) belongs to the 49ers. While there are many teams having nearly the highest curve, the Steelers’ curve is clearly the highest past about 8 YTG. Figure 4 shows the logistic curves for the probability of an average offensive team converting a third down and x yards to go against each of the 32 defensive teams. The teams with the highest and lowest curves are noted on Figure 4.

Table 6: Estimated parameters α_O , α_D , β_O , and β_D for Model 3.

Team	$\alpha_{O,i}$		$\alpha_{D,i}$		$\beta_{O,i}$		$\beta_{D,i}$	
	Posterior	Posterior	Posterior	Posterior	Posterior	Posterior	Posterior	Posterior
	Mean	StDev	Mean	StDev	Mean	StDev	Mean	StDev
49ers	0.4918	0.1573	-0.0494	0.1334	-0.1751	0.0246	0.0081	0.0229
Bears	0.6404	0.1509	0.1000	0.1360	-0.1863	0.0235	0.0174	0.0232
Bengals	0.7378	0.1494	0.0272	0.1344	-0.1300	0.0238	-0.0052	0.0240
Bills	0.5012	0.1567	-0.0391	0.1376	-0.1645	0.0255	-0.0049	0.0225
Broncos	0.6300	0.1539	-0.0413	0.1350	-0.1450	0.0233	-0.0174	0.0227
Browns	0.7418	0.1543	0.1331	0.1396	-0.1468	0.0238	-0.0092	0.0224
Buccaneers	0.6491	0.1487	-0.0238	0.1356	-0.1745	0.0247	0.0113	0.0230
Cardinals	0.6859	0.1540	-0.0663	0.1357	-0.1708	0.0235	-0.0018	0.0219
Chargers	0.5937	0.1543	0.0025	0.1358	-0.1327	0.0231	0.0002	0.0228
Chiefs	0.5584	0.1515	0.1094	0.1401	-0.1738	0.0236	0.0260	0.0232
Colts	0.8214	0.1506	-0.1587	0.1411	-0.1343	0.0243	0.0257	0.0244
Cowboys	0.7334	0.1547	-0.0387	0.1376	-0.1441	0.0232	-0.0091	0.0217
Dolphins	0.6862	0.1516	-0.0638	0.1368	-0.1575	0.0236	-0.0177	0.0220
Eagles	0.8093	0.1511	0.0346	0.1343	-0.1827	0.0246	0.0101	0.0224
Falcons	0.5839	0.1515	0.0363	0.1353	-0.1642	0.0227	0.0234	0.0246
Giants	0.8166	0.1526	0.0559	0.1392	-0.1782	0.0247	-0.0041	0.0221
Jaguars	0.6307	0.1495	-0.0088	0.1377	-0.1192	0.0225	-0.0156	0.0215
Jets	0.5630	0.1507	-0.0750	0.1378	-0.1370	0.0231	0.0282	0.0242
Lions	0.6039	0.1533	0.0285	0.1331	-0.1657	0.0241	-0.0439	0.0225
Packers	0.6030	0.1506	0.0815	0.1372	-0.1328	0.0241	-0.0144	0.0218
Panthers	0.5995	0.1495	-0.0113	0.1345	-0.1715	0.0241	-0.0180	0.0220
Patriots	0.8344	0.1552	0.0896	0.1390	-0.1458	0.0250	0.0244	0.0236
Raiders	0.6570	0.1525	-0.0122	0.1360	-0.1540	0.0237	0.0233	0.0222
Rams	0.5706	0.1521	-0.0254	0.1393	-0.1515	0.0224	-0.0142	0.0217
Ravens	0.6084	0.1500	0.0641	0.1363	-0.1685	0.0237	0.0198	0.0230
Redskins	0.6266	0.1501	-0.0135	0.1348	-0.1513	0.0241	0.0209	0.0231
Saints	0.7252	0.1466	0.0031	0.1371	-0.1421	0.0244	-0.0288	0.0220
Seahawks	0.5531	0.1527	0.0074	0.1356	-0.1675	0.0247	0.0002	0.0218
Steelers	0.7427	0.1518	-0.0090	0.1369	-0.1197	0.0213	-0.0082	0.0214
Texans	0.7700	0.1515	-0.0397	0.1383	-0.1800	0.0263	-0.0041	0.0226
Titans	0.7068	0.1523	-0.0920	0.1349	-0.1567	0.0243	0.0298	0.0230
Vikings	0.6563	0.1539	0.0208	0.1357	-0.1669	0.0236	-0.0175	0.0219

The Colts were the weakest third-and-1 team, while the Lions were the worst third-and-long team. (We consider third down and six or more yards to go to be third and long.) The Chiefs and Patriots are the two best teams at stopping third and long.

From Table 6, we can estimate the probability of any team successfully converting a third down and x YTG against any defensive team, under the assumptions of Model 3. Consider, for example the probability of the 49ers converting a third-and- x against the Chiefs (the best at stopping third-and-1) and the Colts (the worst at converting a third-and-1). The estimated logistic curve for the 49ers versus the Chiefs is

$$\begin{aligned}\pi(x) &= \frac{\exp\left[(\hat{\alpha}_{O,1} - \hat{\alpha}_{D,10}) + x(\hat{\beta}_{O,1} - \hat{\beta}_{D,10})\right]}{1 + \exp\left[(\hat{\alpha}_{O,1} - \hat{\alpha}_{D,10}) + x(\hat{\beta}_{O,1} - \hat{\beta}_{D,10})\right]} \\ &= \frac{\exp[(0.4918 - 0.1094) + x(-0.1751 + 0.0260)]}{1 + \exp[(0.4918 - 0.1094) + x(-0.1751 + 0.0260)]} \\ &= \frac{\exp(0.3824 - 0.1491x)}{1 + \exp(0.3824 - 0.1491x)}.\end{aligned}$$

The curve for the 49ers versus the Colts is determined similarly. The probability of the 49ers converting a third-and-1 is 0.545 against the Chiefs and 0.611 against the Colts. These two curves, displayed in Figure 5, show the extent of the differences among the offensive teams. Figure 6 shows how the curves can differ for the 49ers (the worst third-and-10 team) against the best (Chiefs) and worst (Lions) at stopping a third-and-10. The probabilities of the 49ers converting a third-and-10 against the Chiefs is 0.164 and against the Lions is 0.300.

For Model 3, Figure 7 shows the probabilities and margins of error for making a third-and-1 and a third-and-10 conversion for all 32 teams, ranked from lowest to highest. Playoff teams for the year 2007 are shown in red.

6. Discussion

To get a rough measure of the logistic fit, consider Figure 8, where we have plotted the actual conversion rates for third-and- x for $x = 1, 2, \dots, 10$ for each of the 32 NFL teams. The curve shown on each graph is the logistic curve if the team played against an average defense. Note that this is not simply the logistic curve estimated from the given team's data. For some teams, such as the Patriots, most of the points are above the curve, suggesting that they tended to play below average defensive teams. Other teams, such as

Figure 5: Estimated probability of conversion for the worst 3rd-and-1 team (the 49ers) versus the best (Chiefs) and worst (Colts) 3rd-and-1 defenses.

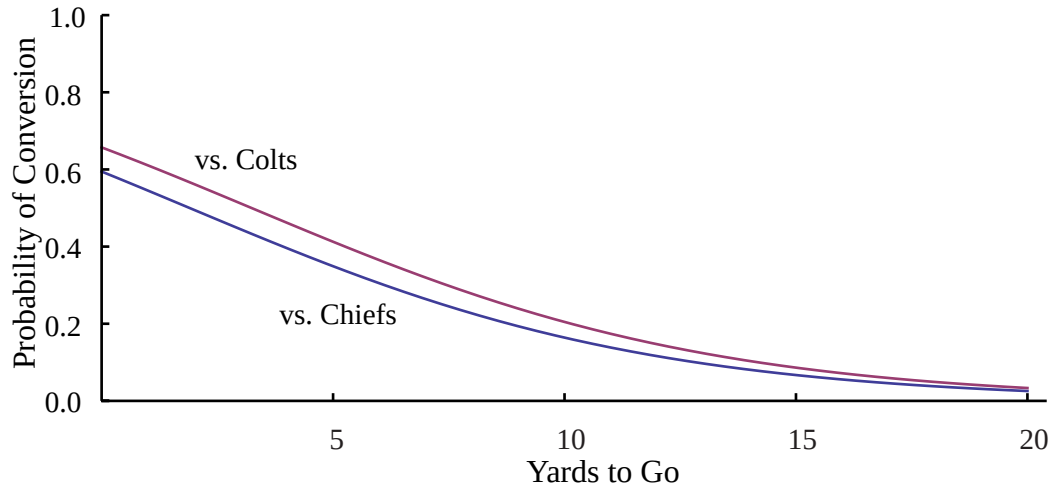


Figure 6: Estimated probability of conversion for the worst 3rd-and-10 team (the 49ers) versus the best (Chiefs) and worst (Lions) 3rd-and-10 defenses.

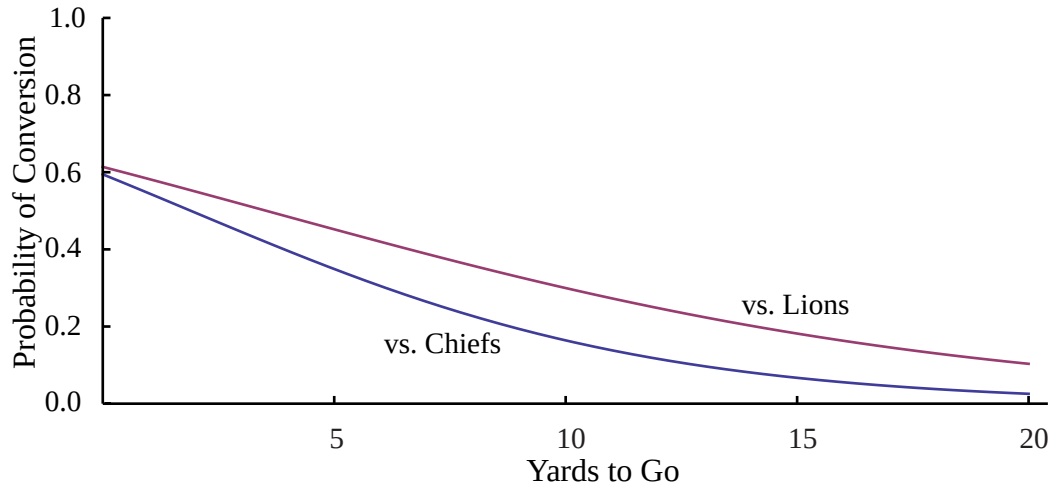


Figure 7. Posterior means and 2.5 and 97.5 percentiles for probabilities of making Third-and-1 and Third-and-10 under Model 3.

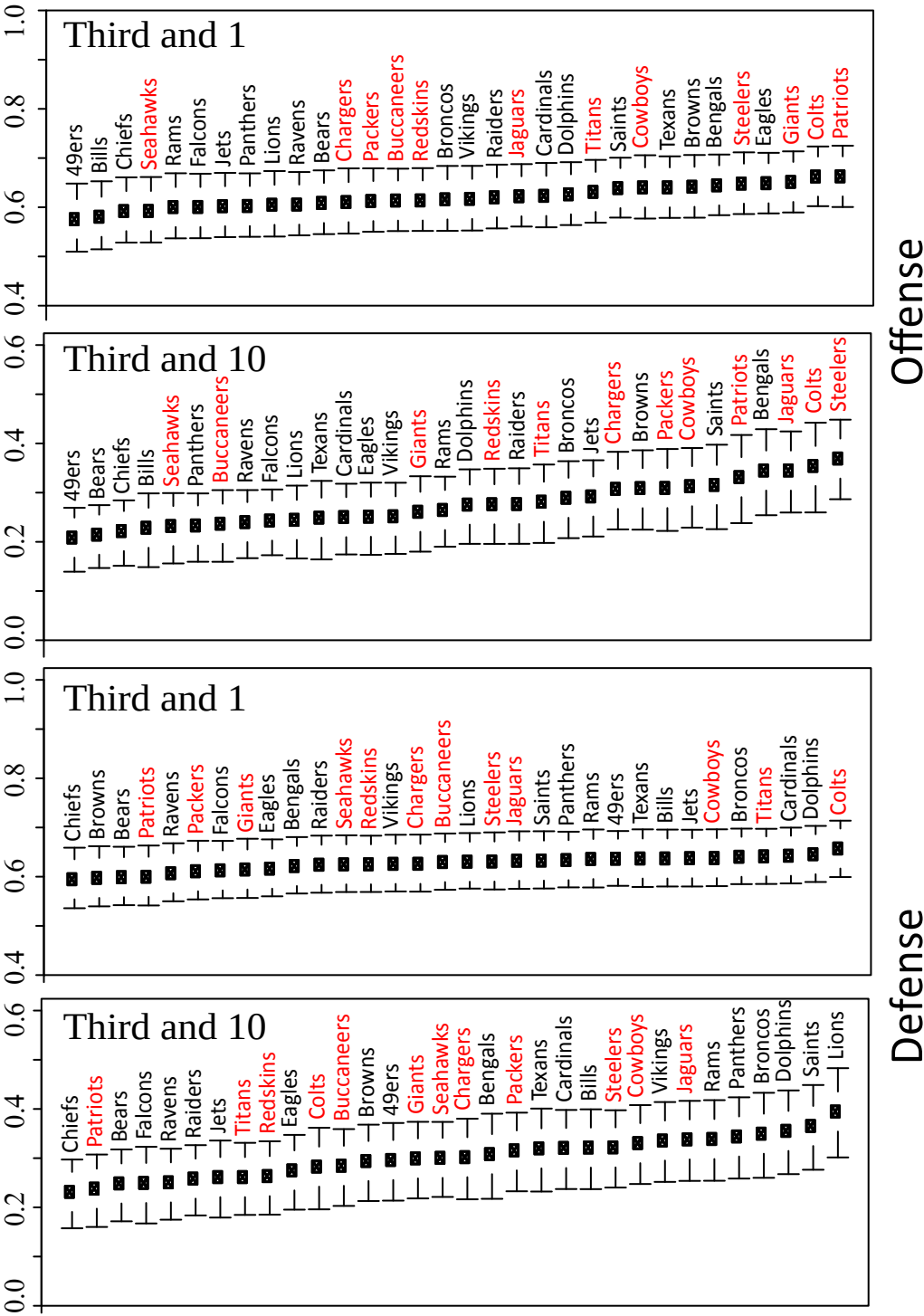
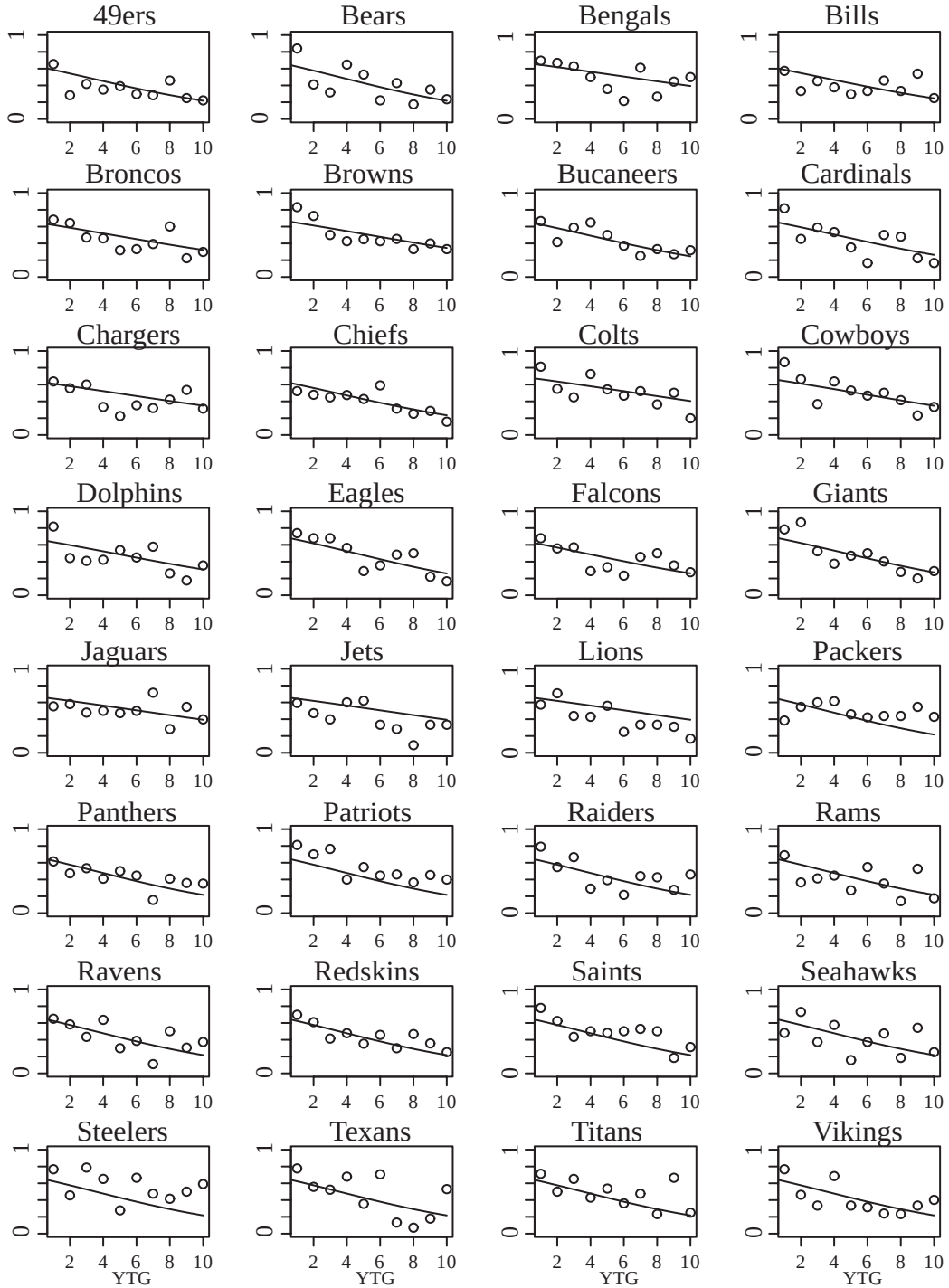


Figure 8. Estimated logistic curves for each offense against an average defense. Circles indicate the actual proportion of successful third-and-x-yards-to-go, for $x = 1, 2, \dots, 10$.



the Jaguars have most points below the curve, suggesting that they tended to play above average defensive teams. The fit of the curves seems fairly good, with the possible exception of third-and-one. Twenty four of the 32 NFL teams had actual third-and-one conversion rates that were above the values predicted (when playing against an average defense). This suggests that third-and-short is slightly easier than the model predicts. It also may be accounted for by the fact that “third-and-inches” is recorded as third-and-one, whereas if more than one yard but less than two yards are needed, the play is recorded as third-and-two. There is a large proportional difference between third-and-inches being recorded as third-and-one, and third-and-just-over-one being recorded as third-and-two.

Comparing different models with different numbers of parameters is inherently difficult, because we would expect models with more parameters to fit better than models with fewer parameters, and to make matters worse, it is often difficult to count parameters in a hierarchical model. The deviance information criterion (DIC) suggested by Spiegelhalter et al. (2002) attempts to account for the “effective number of parameters” in a model, as well as the fit of the model. The DIC is the Bayesian analogue to the Akaike information criterion. Define

$$\begin{aligned}
 D(y, \theta) &= -2 \log p(y|\theta) && \text{the deviance} \\
 \hat{D} &= D(y, \hat{\theta}(y)) && \text{where } \hat{\theta}(y) \text{ is the posterior mean of } \theta \text{ given } y \\
 \bar{D} &= \frac{1}{M} \sum_{m=1}^M D(y^{(m)}, \theta^{(m)}) && \text{the average deviance across } M \text{ simulations} \\
 p_D &= \bar{D} - \hat{D} && \text{a measure of the effective number of parameters} \\
 \text{DIC} &= 2\bar{D} - \hat{D} = \hat{D} + 2p_D && \text{the deviance information criterion.}
 \end{aligned}$$

Smaller values of DIC indicate a better fitting model, adjusted for the number of parameters. The values of DIC, calculated in WinBUGS, are shown in Table 7. Model 1, where all teams are assumed to have the same logistic curve has the lowest DIC, mainly because of the small number of effective parameters (2.00). Model 3, with slope and intercept parameters for both the offensive and defensive teams, has the highest DIC. The results in Table 7 suggest that third-down conversion rates could be the same, or nearly so, across all NFL teams. This casts doubt on the effectiveness of using third-down conversion rates, even adjusted for yards to go, to measure the strength of an offense.

Table 7: Deviance Information Criterion (DIC) for the three models

Model	Parameters	$\bar{D}$	$\hat{D}$	p_D	DIC
1	α, β	8638.89	8636.89	2.00	8640.88
2	α_{Oi}, β_{Oi} (offensive team)	8607.72	8572.36	35.36	8643.07
3	α_{Oi}, β_{Oi} (offensive team) α_{Dj}, β_{Dj} (defensive team)	8584.80	8517.16	67.64	8652.45

7. Summary and Conclusion

We have proposed three models for estimating the probability of converting a third down play. The simplest model assumes that the probability of conversion depends on YTG, but not on the offensive or defensive team. The most complicated is one that depends on (1) the number of yards to go, (2) the strength of the offense, and (3) the strength of the defense. A hierarchical Bayes method is used to avoid the identifiability issue that arises when we look at the difference between the strength of the offense and the strength of the defense. Gibbs sampling is used, through the software package WinBUGS, to estimate the posterior distributions of the parameters.

The raw third down conversion rate is a practically meaningless measure of offensive efficiency since it is so heavily dependent on YTG. The conversion probability is estimated to be a decreasing function of yards to go, but the curves differ little across all 32 teams. In fact, the model with the lowest DIC is the model where the logistic curve is assumed to be the same for all teams.

A further analysis of third down efficiency might focus on field position as well as play selection. For example, teams may be less aggressive in trying for a first down when they are already in field goal range. Also, there are times, such as third-and-20, where a team will virtually give up on getting a first down by selecting a running play (which has little chance of making a first down) rather than a pass play (which carries a fair chance of making a first down but also carries risk, such as a sack or interception).

Appendix A: Convergence of MCMC

We applied WinBUGS to Model 3 with data from the 2007 NFL season using four sets of initial values of the parameters and ran 3000 iterations of the Markov chain (which takes about thirty minutes on a PC running Windows).

After these 3000 iterations, we checked the trace plots and the BGR plots (based on the work of Gelman and Rubin (1992) and Brooks and Gelman (1998)). The trace plots of the various parameters (for all four starting values) suggest thorough mixing within a hundred or so iterations. The BGR plots, which compare the values of the posterior draws from the various starting values, suggest thorough mixing for most parameters after a thousand or so iterations. For some parameters, the BGR plots do not stabilize until about 2500 iterations. To be rather confident of convergence, we take the burn-in period to be the first 10,000 iterations. We then ran through 110,000 iterations from each of four initial values for the parameters, discarding the first 10,000 from each stream. The results are then based on these $4 \times 100,000 = 400,000$ steps through the Markov chain.

Appendix B: Choice of Prior Distributions

We have tried to choose prior distributions for Model 3 that are reasonable for the type of problem considered here. With the hierarchical Bayes method for Model 3, we allow the prior distributions of the model parameters $(\alpha_O, \alpha_D, \beta_O, \beta_D)$ to depend on unknown parameters called hyperparameters. The choice of a prior for the hyperparameters has less influence than the choice of a prior for a model parameter.

Consider first the parameters α_{Oi} , $i = 1, 2, \dots, 32$. We assume that the prior is $N(\mu_1, \tau_1^{-1})$, where μ_1 and τ_1 are the (unknown) hyperparameters. By contrast, the parameters α_{Dj} , $j = 1, 2, \dots, 32$ are centered at zero to avoid the identifiability problem. Consider an offense playing against an average defense, that is, a defense having $\alpha_{Dj} = 0$. For a “third-and- ε ” play, where ε is a number very close to zero, the probability of a successful conversion is

$$\pi = \frac{\exp [\alpha_{Oi} - \alpha_{Dj} + \varepsilon (\beta_{Oi} - \beta_{Dj})]}{1 + \exp [\alpha_{Oi} - \alpha_{Dj} + \varepsilon (\beta_{Oi} - \beta_{Dj})]} \approx \frac{\exp [\alpha_{Oi}]}{1 + \exp [\alpha_{Oi}]}.$$

A reasonable estimate of this probability, *a priori*, is about $2/3$; for this estimate we have used data from Rigdon and Rigdon (2004) based on the 2003 NFL season. With $\pi = 2/3$, we find that

$$\alpha_{Oi} = \log \frac{\pi}{1 - \pi} = \log \frac{2/3}{1/2} = \log 2 \approx 0.693.$$

We thus take 0.693 as the center for the distribution of μ_1 ; note that we are not taking 0.693 as the center for the prior distribution of α_{Oi} (where the mean is μ_1), but rather for the center of the prior for μ_1 . For the precision of the

prior for μ_1 , we take the center to be 25, that is, with a standard deviation of $\frac{1}{5} = 0.2$. This implies that the range from

$$0.693 - 2(0.2) = 0.293 \quad \text{to} \quad 0.693 + 2(0.2) = 1.093$$

should cover most of our prior beliefs for μ_1 . The prior for μ_1 is therefore taken to be $N(0.693, \frac{1}{25})$.

Now consider the prior for τ_1 . A team that converted 80% of third-and- ε would be considered excellent; for such a team

$$\alpha_O = \log \frac{0.8}{1 - 0.8} = \log 4 \approx 1.386.$$

On the other hand a team that made only 50% of third-and- ε would be considered poor, and for such a team we would have

$$\alpha_O = \log \frac{0.5}{1 - 0.5} = 0.$$

Thus we would expect the α_O 's to be between 0 and 1.386 (centered at about 0.7), and if this is the center plus and minus 2 standard deviations, then a reasonable value for the standard deviation is $0.7/2 = 0.35$. The precision should thus be about $\tau_1 = 1/0.35^2 \approx 8$. We therefore take the prior to be centered at 8 with a fairly large variance so as to make the prior somewhat noninformative. Since the precision must be a positive quantity, we choose the prior to be $\text{GAMMA}(1, 0.125)$; in WinBUGS, the $\text{GAMMA}(\alpha, \beta)$ distribution is parameterized to have mean α/β and variance α/β^2 .

The prior distribution for α_D is $N(0, \tau_2^{-1})$, where τ_2 is the only hyperparameter. We choose the same prior distribution for τ_2 as we did for τ_1 , figuring that our prior knowledge about the α_D 's will be about as spread out as our knowledge of the α_O 's.

Our prior for β_O is taken to be $N(\mu_2, \tau_3)$, so we now try to find reasonable priors for the hyperparameters μ_2 and τ_3 . The β_O parameter is like a slope parameter, so we cannot elicit prior information about its prior solely from third-and- ε situations. Instead, let's consider the third-and-10 situation. For average teams $\alpha_O \approx 0.693$ and $\alpha_D = \beta_D = 0$, so we would have

$$\pi = \frac{\exp [(\alpha_{Oi} - \alpha_{Dj}) + 10 (\beta_{Oi} - \beta_{Dj})]}{1 + \exp [(\alpha_{Oi} - \alpha_{Dj}) + 10 (\beta_{Oi} - \beta_{Dj})]} \approx \frac{\exp [0.693 + 10\beta_{Oi}]}{1 + \exp [0.693 + 10\beta_{Oi}]}.$$

Solving for β_O gives

$$\beta_O = \frac{1}{10} \left(\log \frac{\pi}{1 - \pi} - 0.693 \right).$$

Consider a low, middle, and high value for the probability π of making a third-and-10.

$$\text{Low: } \pi = 0.2, \text{ leading to } \beta_O = \frac{1}{10} \left(\log \frac{0.2}{1 - 0.2} - 0.693 \right) \approx -0.208$$

$$\text{Middle: } \pi = 0.35, \text{ leading to } \beta_O = \frac{1}{10} \left(\log \frac{0.35}{1 - 0.35} - 0.693 \right) \approx -0.131$$

$$\text{High: } \pi = 0.5, \text{ leading to } \beta_O = \frac{1}{10} \left(\log \frac{0.5}{1 - 0.5} - 0.693 \right) \approx -0.0693.$$

We take the prior mean for μ_2 to be near the center -0.131 . If the high and low values are two standard deviations above and below the mean, then the standard deviation would be about 0.03, making the precision about 1000. We take the prior for μ_2 to be $N(-0.131, \frac{1}{1000})$.

Consider next, the prior for the hyperparameter τ_3 . Since β_O is a “slope” parameter, we consider low and high slopes:

$$\text{Low: 3rd-and-}\varepsilon \text{ } (\pi = 0.5), \text{ 3rd-and-10 } (\pi = 0.4), \text{ leading to } \beta_O \approx -0.041$$

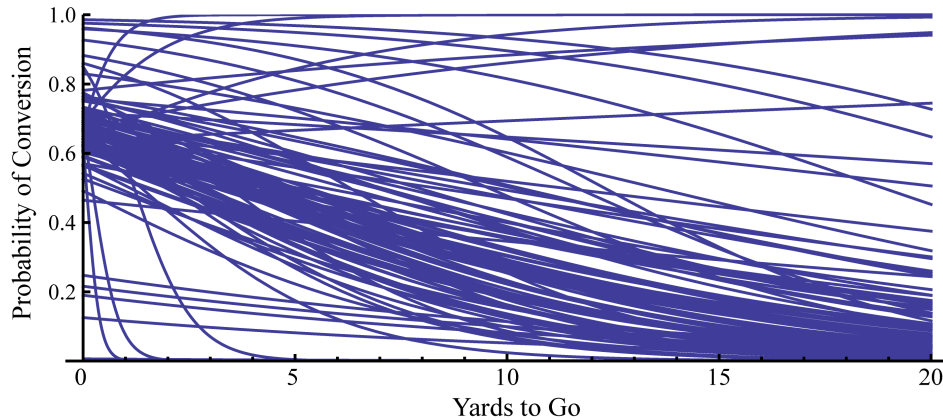
$$\text{High: 3rd-and-}\varepsilon \text{ } (\pi = 0.9), \text{ 3rd-and-10 } (\pi = 0.1), \text{ leading to } \beta_O \approx -0.439.$$

If these are two standard deviations below and above the center, then the standard deviation should be about 0.0995, making the precision approximately 100. Our prior for τ_3 is then taken to be $\text{GAMMA}(1, 0.01)$. The prior for τ_4 is the same.

Finally, we can assess our choice of prior distributions by looking at what kinds of logistic curves we obtain by selecting from the various priors. Figure 9 shows the results of simulating 100 sets of parameters, beginning with the hyperparameters and finishing with the model parameters. We want to guard against choosing priors that are too tight, that is, have too little variation, which can cause the posteriors to be highly dependent on where we choose the priors. In Figure 9, we see a wide variety of logistic curves. Although some of these curves are totally unrealistic, it is hard to imagine that this set of prior distributions would exclude any reasonable logistic curve for third down conversion. This suggests that our priors are sufficiently diffuse, and therefore

reasonably noninformative. Therefore, the posteriors will be dominated by the data, not the prior.

Figure 9: One hundred logistic curves whose parameters were selected at random using the prior and hyperprior distributions.



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